databricks

Advance Lakeflow Jobs Features

LECTURE

Lakeflow Jobs in Production

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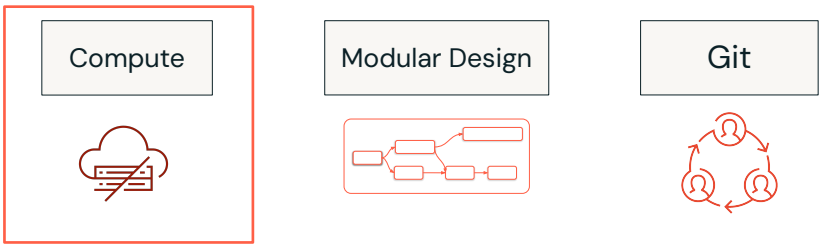
Moving from development to production requires understanding how to design, deploy, and operate Lakeflow Jobs at enterprise scale with appropriate governance, security, and operational practices.

Lakeflow Jobs in Production

Common Best Practices



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Production deployments require attention to four critical areas:

Compute Strategy: Selecting the right compute options for performance, cost, and operational requirements.

Modular Design: Implementing architecture patterns that support maintainability, reusability, and team collaboration.

Git Integration: Version control and deployment practices that ensure consistency and enable CI/CD workflows.

Performance Monitoring: Proactive monitoring and optimization strategies that ensure SLA compliance and cost efficiency.

Lakeflow Jobs in Production

Selecting Compute

Interactive Clusters

It is best for performing ad-hoc analysis, data exploration, or development, **but not production**

- Costly for Job runs
- Limited Scalability
- Availability could be an issue because of parallel usage

Job Clusters

It is **cheaper** as they terminate when the job ends, reducing resource usage and costs. But,

- Start-up Latency
- Subject to cloud provider start-up time
- Maintenance Burden

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Understanding compute options is crucial for production success:

Interactive Clusters are ideal for development and ad-hoc analysis but present significant challenges for production use:

- Cost Issues: Interactive clusters remain running even when not executing jobs, leading to unnecessary costs
- Limited Scalability: Shared resources can create contention and performance unpredictability
- Availability Concerns: Multiple users sharing clusters can cause resource conflicts and job delays

Job Clusters offer better production characteristics:

- Cost Efficiency: Clusters terminate when jobs complete, eliminating idle resource costs
- Dedicated Resources: Each job gets dedicated compute resources, ensuring predictable performance
- Startup Latency: Cloud provider startup times can delay job execution, requiring consideration in SLA planning
- Maintenance Overhead: Requires more configuration and management compared to serverless options

Lakeflow Jobs in Production

Selecting Compute

Serverless Compute

Simplicity: You no longer need to choose the VM type. Customers do not need dedicated skilled DevOps folks

High efficiency: Photon turned on default

Faster startup: VMs run in Databricks account. We ensure enough nodes are available based on historical data and ML

Reliability: Shielded from cloud disruptions

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Serverless compute represents the optimal choice for most production workloads:

Operational Simplicity: Eliminates the need for infrastructure management, VM type selection, and capacity planning. This reduces the specialized DevOps skills required and allows teams to focus on business logic rather than infrastructure.

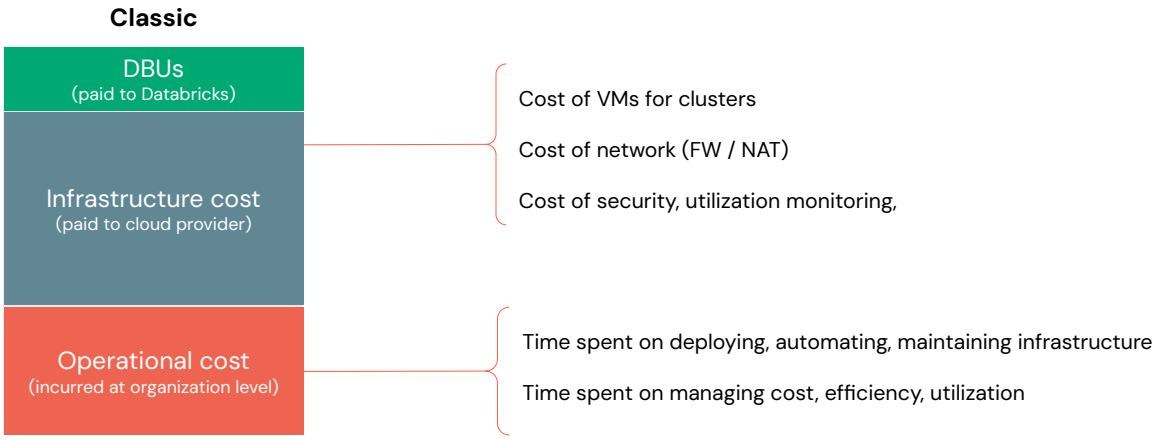
Performance Optimization: Photon acceleration is enabled by default, providing significant performance improvements without additional configuration or cost.

Reliability and Speed: VMs run in Databricks' managed account with predictive scaling based on historical usage patterns and machine learning algorithms, ensuring resources are available when needed.

Cloud Independence: Serverless shields you from cloud provider disruptions and infrastructure issues, providing higher reliability than self-managed clusters.

Lakeflow Jobs in Production

Pricing Structure



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The traditional pricing model involves multiple cost components:

Direct Costs: DBUs paid to Databricks plus infrastructure costs paid directly to cloud providers (VMs, networking, security services).

Operational Overhead: Often overlooked but significant costs including time spent on infrastructure deployment, automation development, maintenance activities, cost monitoring, and efficiency optimization.

Hidden Complexity: Managing multiple billing relationships, optimizing across different cost categories, and maintaining expertise in cloud infrastructure management.

Lakeflow Jobs in Production

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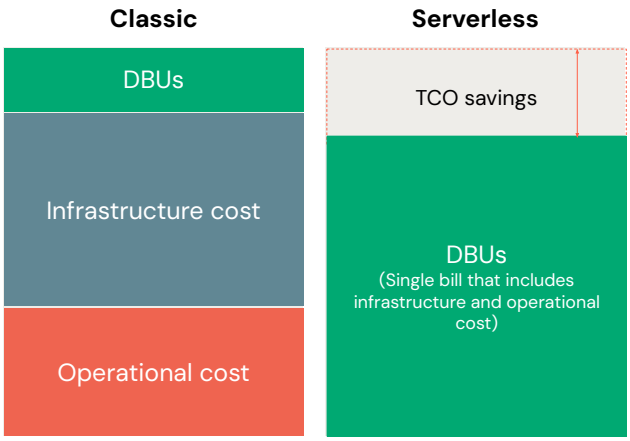

Value

Fully managed service - operationally simpler, more reliable

Fast clusters, auto-scaling - better user experience, lower cost

Out of the box performance and optimizations - lower overall TCO

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Serverless fundamentally simplifies the cost model:

Unified Billing: Single DBU price that includes infrastructure and operational costs, eliminating the need to manage multiple vendor relationships and cost optimization strategies.

Value Proposition: The fully managed service provides operational simplicity and reliability improvements that often justify higher per-unit costs through reduced operational overhead.

Performance Benefits: Auto-scaling capabilities and out-of-the-box optimizations often deliver better performance at lower total cost than self-managed alternatives.

TCO Advantages: When you factor in operational overhead, the total cost of ownership is typically lower with serverless, especially for organizations without dedicated platform engineering teams.

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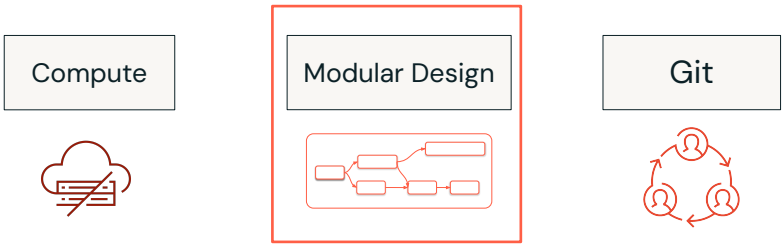

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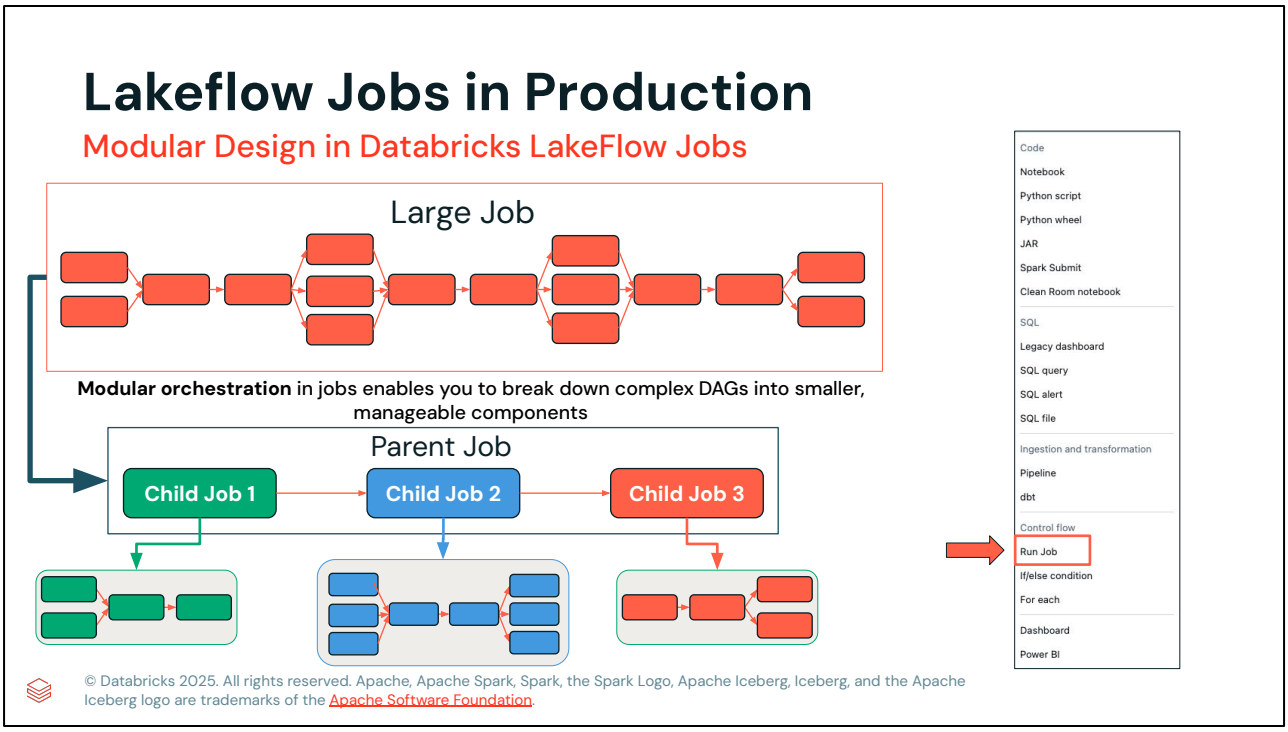

Modular orchestration transforms large, monolithic workflows into maintainable, reusable components:

Decomposition Strategy: Break complex DAGs into logical business units rather than technical components. Each module should represent a cohesive business function that can be developed, tested, and deployed independently.

Parent-Child Relationships: Parent jobs orchestrate child jobs, creating clean separation of concerns while maintaining overall workflow coordination.

Benefits Realization:

- Maintainability: Smaller jobs are easier to understand, modify, and troubleshoot
- Reusability: Child jobs can be reused across multiple parent workflows
- Team Collaboration: Different teams can own different modules while collaborating on the overall workflow
- Testing: Individual modules can be tested independently, improving quality and reducing deployment risk

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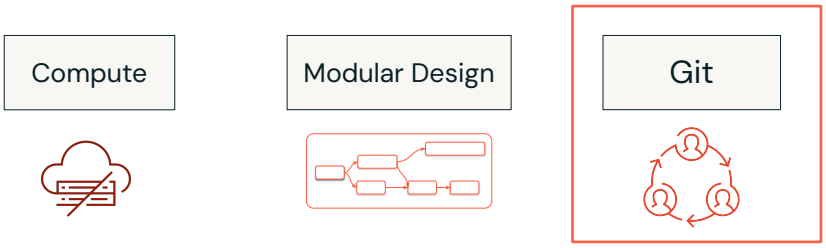
Common Best Practices



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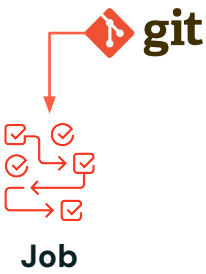
Jobs and Git



Jobs supports execution of notebooks from Git

- This **prevents unintentional changes** to your **production** job, for example, when another user makes local edits in a 'Prod' repo or switches branches
- **Simplifying the job definition** process by having a single source of truth
- Makes **deployment** easy for notebooks through CI/CD
- Users can **connect** with **various Git platforms**, as Databricks supports GitHub, GitLab, AWS CodeCommit, and other Git providers.

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Git integration provides essential capabilities for production workflow management:

Change Management: Prevents unintentional changes to production jobs by ensuring all modifications go through proper version control processes.

Single Source of Truth: Eliminates confusion about which version of code is running in production by always executing from committed code in specific branches or tags.

CI/CD Integration: Enables automated testing and deployment pipelines that can validate changes before they reach production environments.

Platform Support: Broad compatibility with GitHub, GitLab, AWS CodeCommit, and other Git providers ensures you can integrate with existing development workflows regardless of your chosen platform.

Collaboration Benefits: Multiple developers can collaborate on workflow development using standard Git workflows (branches, pull requests, code reviews) while maintaining production stability.

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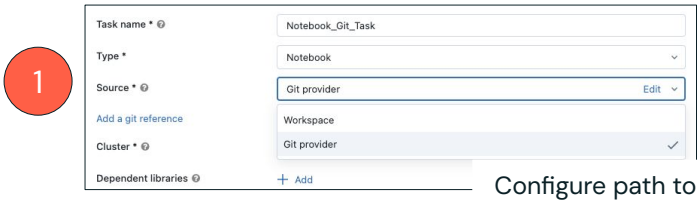
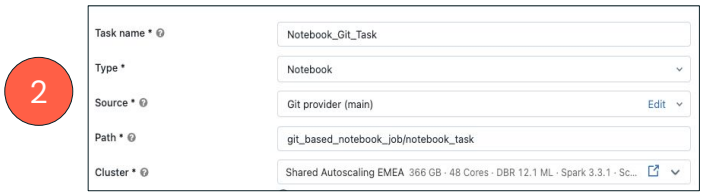
Jobs and Git Configuration Steps

Create task with Git provider as Source



Configure path to main notebook under repository root



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Implementing Git integration involves two straightforward steps:

Step 1 - Source Configuration: Create tasks with Git provider as the source, specifying the repository, branch/tag, and authentication credentials. This ensures your job always executes the committed version of your code.

Step 2 - Path Configuration: Configure the path to your main notebook under the repository root, ensuring the job can locate and execute the correct entry point for your workflow.

Best Practices: Use specific branches or tags for production rather than constantly changing main branches, implement proper code review processes before merging to production branches, and maintain clear documentation about which repositories and paths contain production job code.

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